

Themis: advancing precision oncology through comprehensive molecular subtyping and optimization

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Abstract

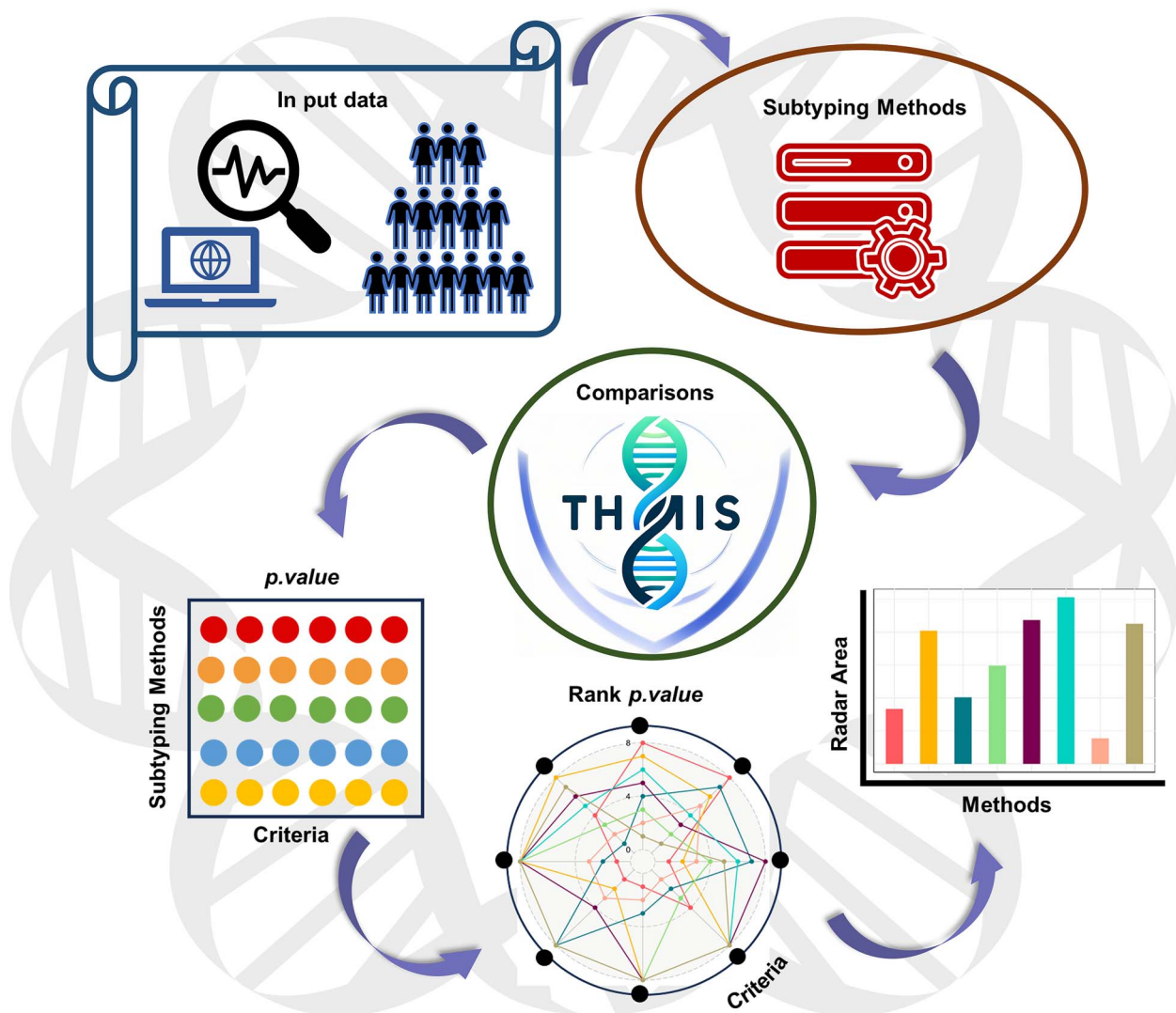
Recent advances in tumor molecular subtyping have revolutionized precision oncology, offering novel avenues for patient-specific treatment strategies. However, a comprehensive and independent comparison of these subtyping methodologies remains unexplored. This study introduces 'Themis' (Tumor Heterogeneity analysis on Molecular subtyping System), an evaluation platform that encapsulates a few representative tumor molecular subtyping methods, including Stemness, Anoikis, Metabolism, and pathway-based classifications, utilizing 38 test datasets curated from The Cancer Genome Atlas (TCGA) and significant studies. Our self-designed quantitative analysis uncovers the relative strengths, limitations, and applicability of each method in different clinical contexts. Crucially, Themis serves as a vital tool in identifying the most appropriate subtyping methods for specific clinical scenarios. It also guides fine-tuning existing subtyping methods to achieve more accurate phenotype-associated results. To demonstrate the practical utility, we apply Themis to a breast cancer dataset, showcasing its efficacy in selecting the most suitable subtyping methods for personalized medicine in various clinical scenarios. This study bridges a crucial gap in cancer research and lays a foundation for future advancements in individualized cancer therapy and patient management.

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Graphical Abstract



Keywords: tumor subtyping; pan-cancer; benchmark testing; bioinformatics

Introduction

Accurate cancer classification, traditionally based on clinical and pathological criteria, has become paramount to precision medicine, now increasingly underpinned by molecular insights for effective diagnosis and tailored oncological decision-making [1]. Historically, cancer classification has hinged upon clinical and pathological parameters, supplemented by a suite of standard biomarkers. However, this conventional approach often overlooks the intricate molecular dynamics of oncogenesis and tumor progression, underscoring the need for advanced molecular subtyping techniques. The advent of high-throughput technologies has precipitated a paradigm shift in cancer classification, unveiling the vast array of genetic alterations and molecular events underlying cancer, thereby significantly expanding our understanding [2, 3]. These advancements have not only unveiled a spectrum of novel diagnostic biomarkers but also have identified potential targets for therapeutic intervention.

The molecular heterogeneity of cancer challenges simple categorization, necessitating a pivot from conventional classification

schemes to those rooted in genomic data. Specifically, classifications derived from gene expression profiles show great potential in customizing treatment protocols, thereby refining prognostic accuracy and enhancing therapeutic outcomes [4–6]. Empirical evidence of this includes Gay et al.'s study, which indicated that patients with the small cell lung cancer (SCLC)-I subtype of SCLC derived substantial benefit from integrating immunotherapy with standard chemotherapy [7]. Thorsson et al. identified six immune subtypes spanning multiple cancer types, predicting differences in overall survival (OS) and progression-free survival (PFS) both within and among different cancers [8]. Wolf et al. used breast cancer subtypes to predict responses to various drugs with different mechanisms of action, presenting a framework for identifying response-predictive subtype patterns to prioritize treatments in the I-SPY2 trial [9, 10].

In recent years, the corpus of research focusing on transcriptomics-driven molecular classification of cancer has expanded markedly, with methodologies broadly divisible into three principal categories: those predicated on feature-associated genes [7, 8, 11–14], those focusing on pathway-related genes

[10, 15–20], and those analyzing patterns of immune cell infiltration within the tumor microenvironment (TME) [21–27]. The gene marker-based classification methods are susceptible to the specificity of cell markers, which can compromise subtyping performances. Although the influences of gene specificity issues may be alleviated in cell infiltration based methods, as they usually rely on deconvolution algorithms [28], the complexity of cancer subtyping performance dramatically increases. Therefore, different subtyping methods may be appropriate for diverse application scenarios.

Despite significant strides in molecular subtyping, the field faces key challenges. Foremost among these is the absence of comprehensive comparative analyses among subtyping methods, which hinders the optimization of these methods for specific datasets and clinical scenarios. Additionally, while individual subtyping approaches have shown distinct advantages, the integration of these diverse methodologies to harness their collective strengths remains unachieved. This gap underscores the pressing need for a critical quantitative assessment of current subtyping techniques and the development of an integrative platform that synergizes the advantages of these diverse methods.

To address these challenges, we have developed Themis (Tumor HETerogeneity analysis on Molecular subtyping System), an evaluation platform designed to benchmark and optimize molecular subtyping methods with the guidance of self-designed quantitative metrics, thereby enhancing precision and applicability in cancer classification. Our evaluation involved eight established subtyping methods, inspecting their robustness, and pan-cancer utility. We categorized these methods according to their operational principles, i.e., pathway-related gene methods [15–17], feature genes approaches [11–13], and deconvolution strategies based on cellular composition [21, 22]. Employing eight testing and 30 pan-cancer datasets from The Cancer Genome Atlas (TCGA), Themis conducted benchmarking, correlating subtyping results with clinical outcomes. This innovative platform not only establishes a benchmarking standard but also guides the creation of novel subtyping methods with enhanced clinical relevance. Ultimately, Themis aims to significantly elevate the precision and practicality of cancer classification, paving the way for more personalized and effective oncological treatments.

Results

Integrating and evaluating diverse subtyping methods: the Themis framework

This study aims to establish a research framework for the comprehensive evaluation of various tumor subtype classification methods. To encompass a broad spectrum of tumor subtyping classification, we collected the datasets and relevant clinical information from the aforementioned literatures and TCGA database (<https://xenabrowser.net/datapages/>) [29] (Figure 1A). Meanwhile, Themis integrates diverse tumor subtyping methods as follows (Figure 1B): Stemness [13], Anoikis [12], Metabolism [11], complemented by two innovative approaches to TME characterization, TME1 [22], TME2 [21], and three pathway-based classifications, Immune-Cancer Pathway [17], Immune Status Pathway [16], and Cancer Pathway [15]. The repeated results of the classification method obtained from the original research were shown in Supplementary Figures 1–11 (Supplementary Information File, Section 4).

Initially, we employed these subtyping methods for analysis of all test datasets, ensuring a thorough evaluation across varying conditions (Figure 1C). Subsequently, we conducted quantitative

analyses to explore correlations between molecular subtyping results and various clinical phenotypes, including OS, PFS, mRNA stemness index (mRNAsi) [30] score, immune infiltration scores (ESTIMATEScore, ImmuneScore, and StromalScore) [31], as well as predictions of immune therapy response (immune checkpoint blockade [ICB]) predicted by TIDE [32] and chemotherapy drug response by pRRophetic [33].

Specifically, our platform's evaluation strategy consists of three components: (1) Robustness: We quantitatively evaluated the performance of each subtyping method on different datasets based on multiple criteria such as OS, PFS, mRNAsi score, immune infiltration score, immunotherapy, and chemotherapy response prediction. (2) Pan-cancer applicability: determined by the conservation of phenotypic associations between different cancer subtypes. Extend the application of each method to pan-cancer data, and quantitatively evaluate and compare their applicability in different cancer types. (3) Openness: Our platform allows users to add custom subtyping methods to compare the applicability of their features in pan-tumor datasets. Furthermore, people can use Themis to evaluate the performance of various subtyping methods on their own dataset. The calculation metrics were shown in Figure 1. The metrics used for this evaluation are extensively outlined in the 'Methods' sections.

Evaluating the robustness across diverse subtyping methods

To rigorously evaluate the robustness of each method in multiple scenarios, these methods were tested on datasets from different studies, and the relationship between the subtyping results and multiple clinical criteria such as OS, PFS, mRNAsi score, immune infiltration score, immunotherapy, and chemotherapy response prediction was quantitatively analyzed. This serves as a standard to evaluate different tumor subtyping methods based on the statistical significance of various metrics, as detailed in the 'Methods' (Supplementary Information File, Section 6).

Figure 2A delineates the significance of our findings, showcasing the results of survival analyses for OS and PFS, and univariate and multivariate cox regression analyses, indicating robust relationships between subtyping results and patient prognoses. Among the methods evaluated, Stemness, Anoikis, and Cancer Pathway, displayed statistical significance ($P < 0.05$). These methods show a high degree of suitability for prognosis-related analyses, indicating their potential in clinical applications. With regards to mRNAsi scores and immune infiltration scores, Stemness, Metabolism, TME1, TME2, Immune-Cancer Pathway, Immune Status Pathway, and Cancer Pathway were appropriate for personalized stratification. Specifically, Immune Status Pathway and Cancer Pathway showed resilience in TIDE immunotherapy prediction across datasets, establishing their robustness in this domain. Moreover, the platform facilitated the selection of optimal subtyping methods for specific datasets. For instance, the Stemness and Immune Status Pathway methods were advantageous for Data1, evidenced by statistically significant outcomes in K-M analysis ($P < 0.05$, Log-rank test), cox regression analysis ($P < 0.05$, Wald test), mRNAsi scores, immune infiltration scores, and immune therapy prediction ($P < 0.05$, ANOVA test). For Data2, except for TME1, other methods were appropriate.

Figure 2B presents an insightful illustration of the performance of various subtyping methods in predicting chemotherapy responses. This analysis, however, excludes Data3 owing to the unavailability of pertinent literature data. For Data1, all methods, except Anoikis, Immune-Cancer Pathway, and Immune Status Pathway, resulted in statistically significant outcomes ($P < 0.05$,

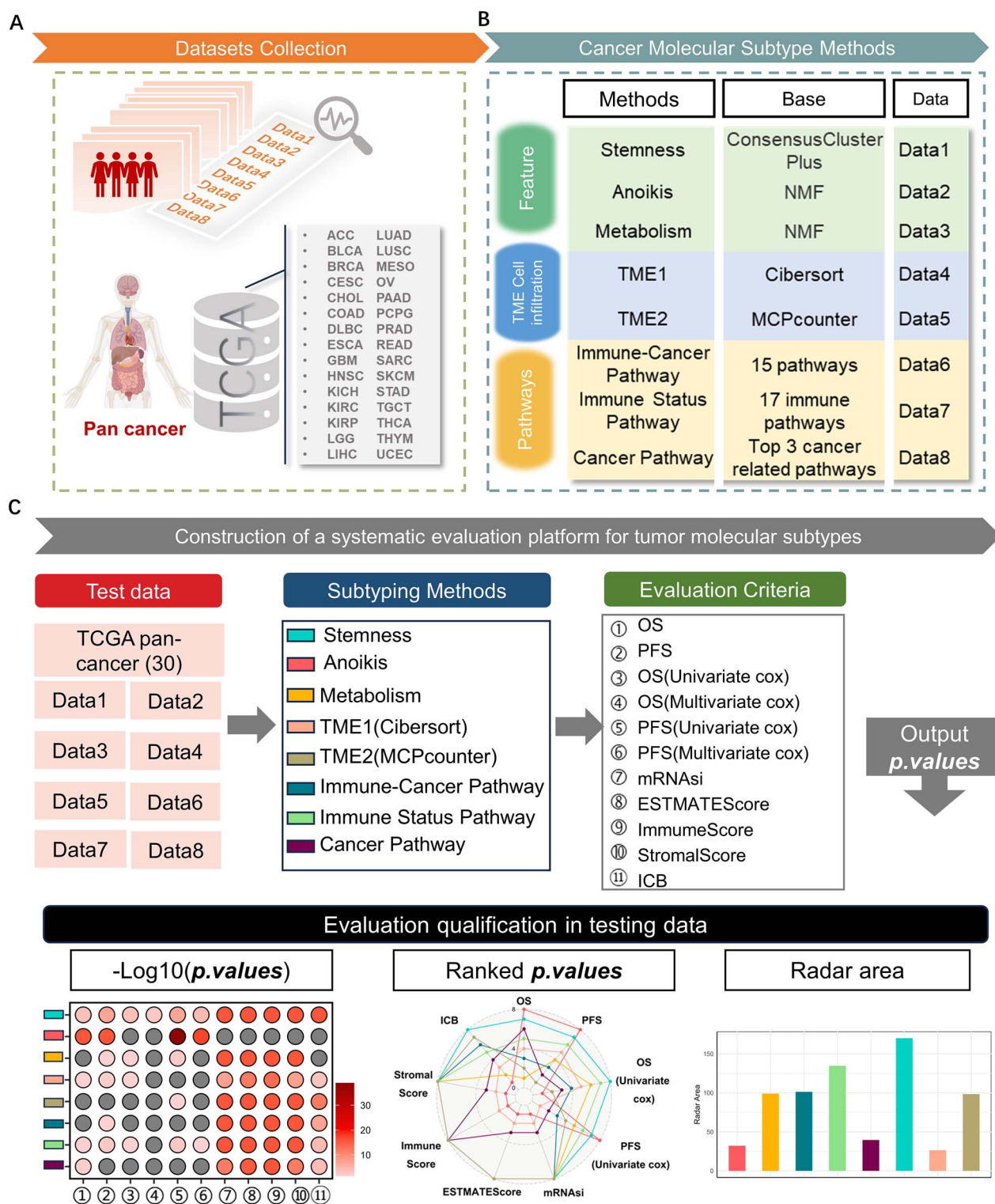


Figure 1. Process design diagram for benchmarking implementation. (A) Finding and downloading information about tumor subtyping methods and their metadata sets, pan-cancer data collection. (B) Datasets from eight tumor subtyping methods from three main categories are benchmarked, with tables showing the basis of subtyping for each method in each typing technique and for the raw data for which it is used. (C) Benchmarking results are measured according to the robustness and usability of the methods, visualized and quantitatively compared.

ANOVA test), with Metabolism standing out. Similarly, in Data2, except for TME1, most subtyping methods showed a comparative advantage, especially TME2, Metabolism, and Cancer Pathway, each yielding statistically significant results across all examined chemotherapy agents.

In the original study, only 27 Anoikis-related genes were used, however, the metadata of Anoikis (Data2) was dynamically changing. To enrich the research, Anoikis gene panel was also obtained through the GeneCards website, and molecular subtyping performances were conducted based on these 912 Anoikis-related

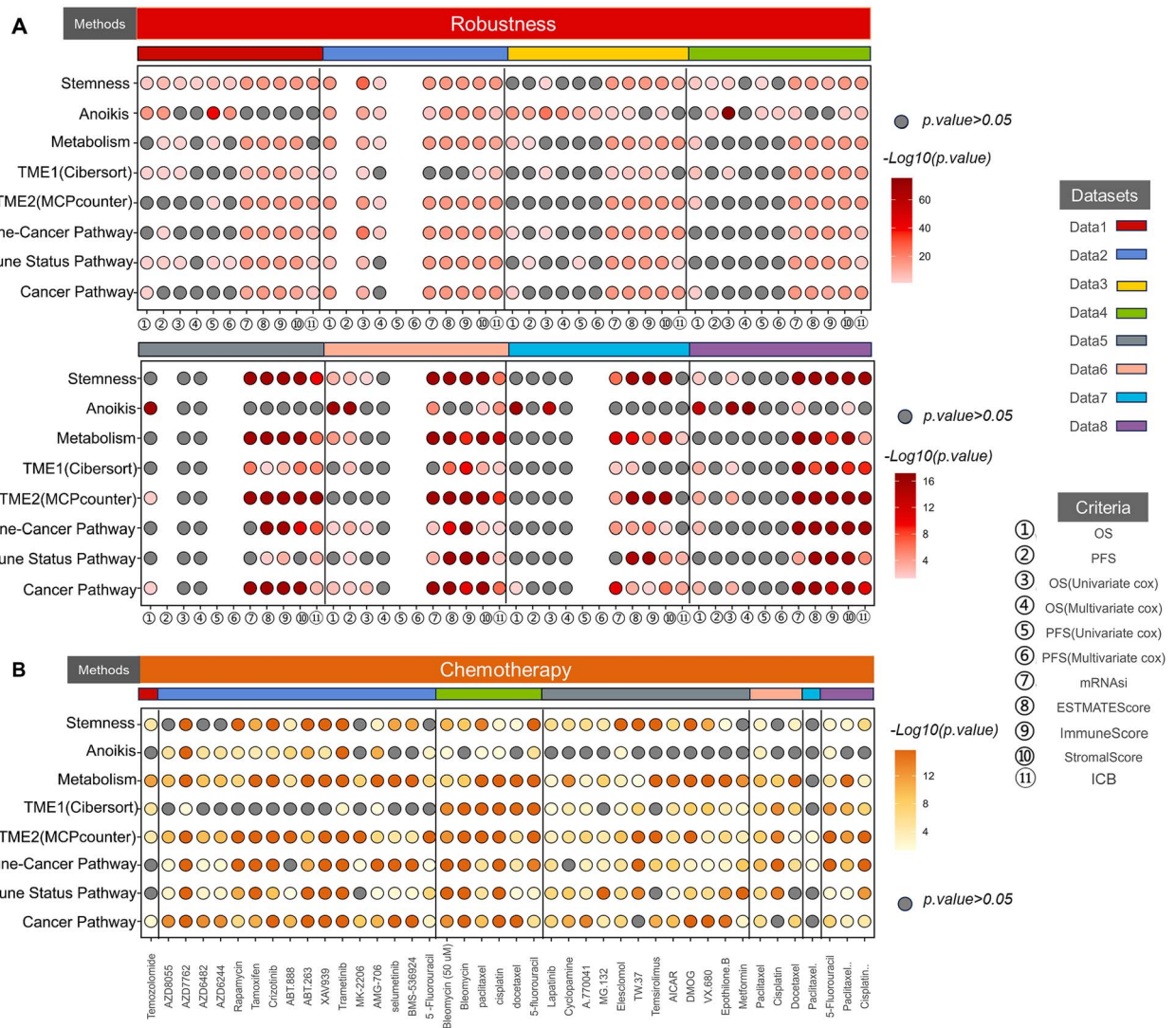


Figure 2. **Summary table of the performance of all methods.** (A) the performance in terms of robustness includes statistical analysis results of K-M analysis (OS/PFS, log-rank test), univariate Cox analysis (OS/PFS, Wald test), mRNasi score, immune infiltration score, and TIDE immunotherapy (ICB) (ANOVA test). (B) the performance in terms of robustness includes statistical analysis results of chemotherapy drugs (ANOVA test). The p values from the comparisons of different classified patient groups in their quantitative results were summarized as the content of the table. We negatively log-transformed p-value ($-\log_{10}(p.value)$) and visualized the results using a bubble chart color coded with p values. If blank with no dots it means that there is no corresponding clinical information available for quantitative calculations for that dataset.

genes. Similarly, the Anoikis (912 genes) method was applied to 38 test datasets to quantitatively analyze the relationship between subtyping results and clinical data. In terms of analyzing survival prognosis, we found that the Anoikis (912 genes) method has statistical significance in multiple datasets (Data1-4, 6, 8). Compared to the Anoikis (27 genes), although the p-value results of Anoikis (27 genes) are more significant, the Anoikis (912 genes) has more statistically significant results (Supplementary Figure 12A). The Anoikis (912 genes) showed good results in mRNasi score, immune infiltration score, and immunotherapy prediction. Compared to the results of Anoikis (27 genes), Anoikis (912 genes) achieved more and greater significant results (Supplementary Figure 12A). In addition, Anoikis (912 genes) has shown a relative advantage in predicting chemotherapy responses among multiple drugs in different datasets (Supplementary Figure 12B).

Ranking of subtyping methods

To further compare the performance of each subtyping method across different datasets, after creating bubble plots visualizing the statistical performance across various phenotypes using normalized p-values, we synthesized comparisons by generating sorted radar charts and calculating their respective areas. The radar charts in Figures 3A and 3B provide a visual representation of the performance rankings of the various subtyping methods across datasets, ordered in descending significance based on p-values. The rankings are derived from nine key metrics, encompassing K-M analysis (OS/PFS), univariate cox analysis (OS/PFS), mRNasi scores, immune infiltration scores, and TIDE immunotherapy (ICB) prediction. Higher rankings, denoting better performance, facilitated a direct comparison of the methods' strengths and weaknesses within the same dataset and across different datasets.

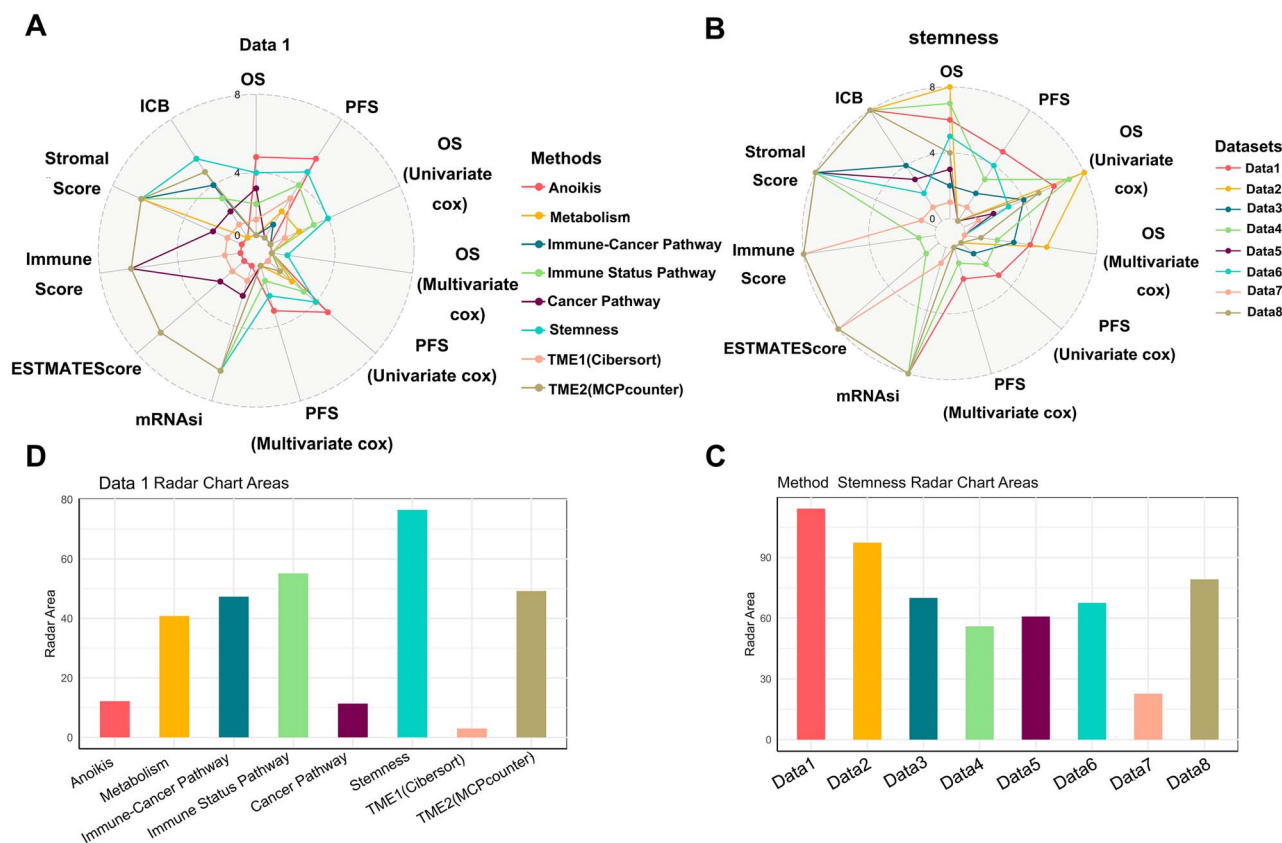


Figure 3. **Performance of all methods for the test datasets.** Proportion of obtained quantitative criterion results with P-values using each dataset and each subtyping method. (A) the radar plot shows quantitative results of correlation with clinical phenotypes using the same dataset (Data1) visualized across all methods. (B) the radar plot shows the quantitative results related to clinical phenotype using the same subtyping method (Stemness) in different data. (C) Bar graphs showing quantitative results of each of the subtyping methods from the Data1 dataset in a quantitative comparison of radar plot areas. (D) Bar graph showing the quantitative comparison of radar plot areas of quantitative results for Stemness methods across all datasets. Different colors correspond to each of the tumor subtyping methods and each of the datasets is indicated in the free plots.

By calculating the total area enclosed within each radar plot (Figure 3C), we were able to determine the most effective subtyping method in relation to clinical phenotypes for each dataset. For instance, the Stemness method prevailed in Data1 across all metrics. Extending this analysis, we compared the performance of individual subtyping methods across various datasets, as depicted in Figure 3D. Take Stemness method as an example, Data1 exhibits the highest performance compared with other datasets. The comparison results of the remaining seven methods are shown in Supplementary Figures 17-20.

Applicability in personalized tumor medicine

To evaluate the pan-cancer applicability of subtyping methods by examining the conservatism of subtype-phenotype associations across different cancer types. We systematically applied each subtyping method to a dataset of 30 different tumor types in TCGA (<https://xenabrowser.net/datapages/>) [29]. This analysis aimed to discern the methods' adaptability and relevance across varied cancers, focusing on molecular heterogeneity and clinical phenotype correlations. The results summarized in bubble plots (Figure 4A), underscore that the selection of an appropriate subtyping method is critically dependent on the inherent tumor heterogeneity and the relevance of clinical phenotypes (see Supplementary Table and 'Methods' section).

Subtyping method selection by users is typically informed by clinical phenotype relevance. Within the TCGA adrenocortical carcinoma (ACC) datasets, for instance, subtyping methods such

as Stemness, Anoikis, Metabolism, TME2, and Cancer Pathway were found to be predictive of survival outcomes, with a majority of these methods yielding statistically significant prognostic quantitative results. For scores related to the tumor mRNasi and microenvironment, subtyping methods incorporating Stemness, Anoikis, Metabolism, TME2, and Immune Status Pathway showed significant correlations with those factors ($P < 0.05$, ANOVA test). However, only Anoikis and TME2 provided statistically significant predictions ($P < 0.05$, ANOVA test) for TIDE immunotherapy outcomes. An integrated analysis of these factors suggests that ACC tumor samples may benefit from employing the Anoikis and TME2 subtyping methods (Figure 4A). This integrative approach allows for the tailored selection of subtyping methods for different tumor types based on specific clinical and molecular criteria.

Further, we exemplified how to employ radar charts to compare the performance of the subtyping methods on ACC datasets. K-M analysis (OS/PFS) indicated the Anoikis method as the most statistically significant ($P < 0.05$, Log-rank test) in ACC. Univariate cox analysis for OS and PFS showed that the Cancer Pathway method was notably significant ($P < 0.05$, Wald test) in ACC (Figure 4B). The comprehensive performance of these subtyping methods, represented by the total radar area encompassing all assessed dimensions, positioned Stemness, Metabolism, and TME2 as the leading methods for ACC, (Figure 4C). The comparison results of these methods in other tumors are shown in Supplementary Figures 21-28.

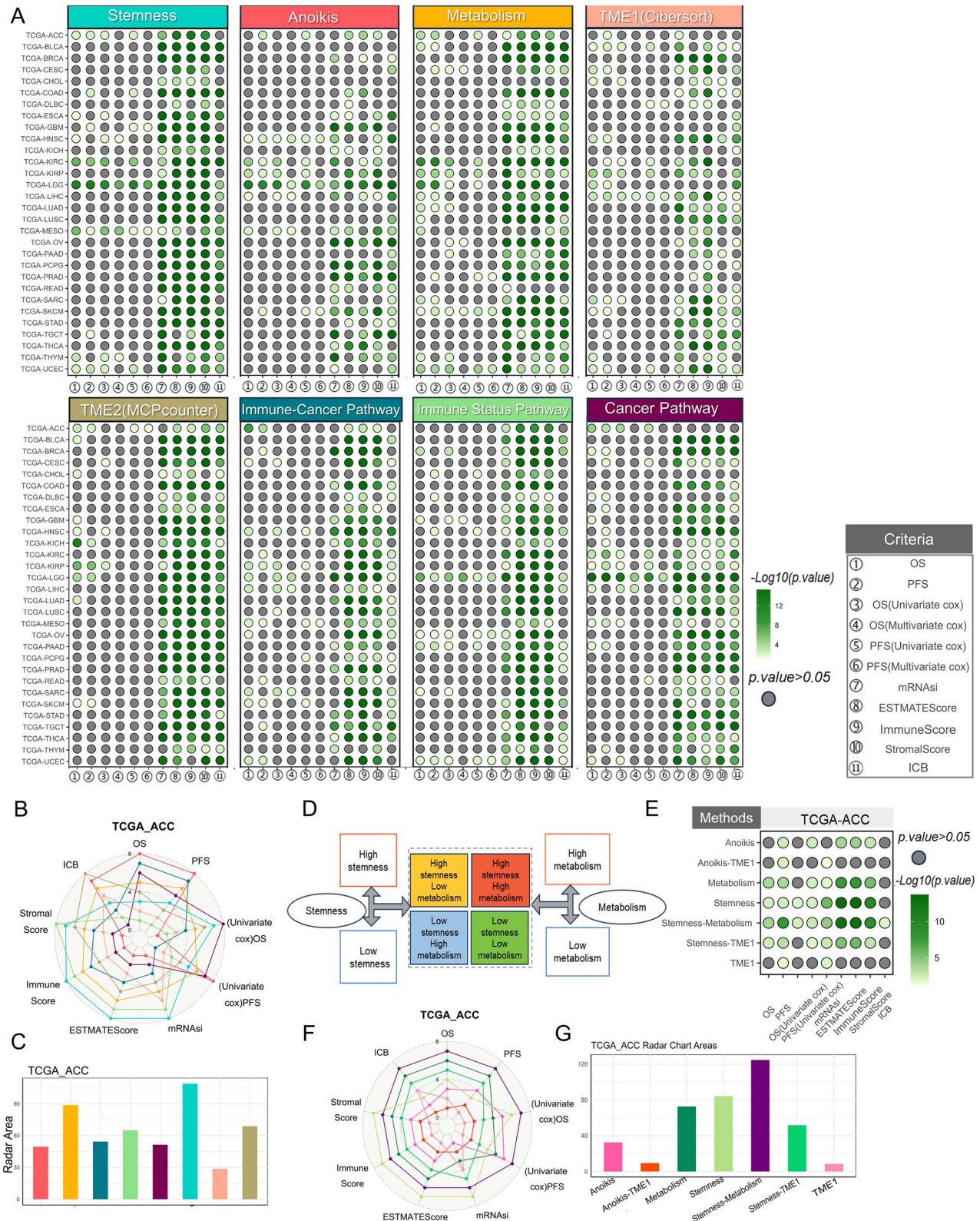


Figure 4. Performance of each subtyping method in pan-cancer. (A) We visualized their performance in terms of pan-cancer availability. (B) Radar plots showing quantitative results of correlation with clinical phenotypes using the ACC dataset visualized for all methods. (C) Bar graph showing the quantitative comparison of radar plot area for quantitative results of methods in the ACC datasets. (D) Optimized results using a combination of two subtyping methods, Stemness, and metabolism (combining high and low stemness with high and low metabolic characteristics). (E) the bubble chart shows the comparative analysis of the performance between the ‘Stemness-metabolism’ method and other methods. (F) the radar plot shows the quantitative results related to clinical phenotype using these different subtyping methods in the same data (ACC). (G) the bar chart displays a quantitative comparison of the radar plot area between the quantitative results of the ‘Stemness-metabolism’ method and other methods in the ACC dataset.

Themis further facilitates the development of tailored subtyping methods. For example, by amalgamating the top-performing methods in ACC, Stemness and Metabolism, we devised a novel subtyping strategy, termed 'Stemness-Metabolism' (detailed in the 'Methods' section, Figure 4D). A subsequent comparative analysis of the 'Stemness-Metabolism' method against the standalone Stemness and Metabolism methods revealed a notable enhancement across several quantitative metrics (Figure 4E). Although the 'Stemness-Metabolism' method yielded a marginally lower score in immune infiltration assessment, compared to the Stemness method, the significance levels of both methods remained high, as evidenced by the highly significant p-values ($P < 0.05$, ANOVA test) obtained, thus confirming the robustness of the results (Figure 4F). Moreover, the Stemness-Metabolism method outperformed the original two methods in survival prognosis and TIDE immunotherapy prediction, significantly enhancing the p values and affirming the reliability of this optimized method. Furthermore, the overall methodological efficacy of the 'Stemness-Metabolism' approach, as measured by the aggregate area under the radar chart, surpassed that of the individual Stemness and Metabolism methods (Figure 4G). However, the optimized result could not be obtained by a random combination of methods. When combining two poor-performing methods, Anoikis and TME1 (termed as Anoikis-TME1); as well as a good-performing method, Stemness, and a poor-performing method, TME1 (termed as Stemness-TME1), inferior results were obtained (Figures 4F, 4G). Such results demonstrate the potential utility of Themis in the guidance of subtyping method design for personalized treatment regimens.

EMT is a crucial biological process in cancer progression, particularly in metastasis. We downloaded all EMT genes from the dbEMT database and termed them as EMT feature-related methods, including the EMT (Consensus) method based on the ConsensusClusterPlus package and the EMT (NMF) method based on the NMF package, respectively, applying them to pan-cancer molecular subtyping studies. The results of the EMT (Consensus) and EMT (NMF) methods show high consistency in most cancer types (Figure 5A). In terms of survival prognosis, both methods demonstrate superior statistical results in seven types of cancer, including ACC, KIRC, KIRP, LGG, and LUAD. Moreover, in terms of tumor mRNAsi, microenvironment-related scores, and prediction of immunotherapy, they also show consistent and statistically significant results in 15 types of cancer.

Comparing these two EMT-related subtyping methods with eight existing subtyping methods, the results from radar charts (Supplementary Figure 15-16) and radar chart area calculations (Figure 5B) show that, considering multiple clinically relevant phenotypes such as comprehensive survival prognosis, tumor immune infiltration scores, stemness status, and immunotherapy prediction, EMT-based subtyping methods perform best in ACC, BLCA, COAD, GBM, LGG, LUSC, OVCA, PRAD, STAD, and UCEC (Figure 5B). This implies that EMT may play an important role in these cancers, and research on EMT related progress may be of importance for understanding the biological characteristics and prognosis of tumors.

Similarly, the Anoikis (912 genes) method was systematically applied to a dataset of 30 different tumor types in TCGA, and its applicability and clinical phenotype correlation were evaluated in various cancers. In the outcome of survival prognosis, it was found that Anoikis (912 genes) had statistical significance in K-M and Cox analysis of six types of tumors (Supplementary Figure 13A). In terms of tumor mRNAsi, microenvironment-related scores, and immunotherapy prediction, Anoikis (912 genes) performed

well in 13 types of tumors with significant statistical significance. Compared to the Anoikis (27 genes) method, Anoikis (912 genes) has higher reliability in various cancers (Supplementary Figure 13A). In addition, radar charts (Supplementary Figure 14A) and their areas (Supplementary Figure 13B) indicate that among the other 24 tumor types, except for CESC, ESCA, KIRC, LIHC, LUAD, and SKCM, the overall performance of the Anoikis (912 genes) is better than that of the Anoikis (27 genes) method.

Demonstrating Themis' impact on personalized breast cancer treatment

Themis is designed as a versatile tool, enabling the selection of optimal subtyping methods tailored to specific datasets and facilitating the performance evaluation of emerging subtyping methodologies. The platform allows users to input their subtyping methods, select data, and employ methods for analyzing test datasets, applying quantitative metrics to benchmark different subtyping methodologies.

As a case study, we applied Denise et al.'s [10] breast cancer subtyping method to our platform, comparing it against established methodologies to determine its suitability for a specific dataset [10] (Supplementary Information File, Section 3). Using the subtyping method outlined in the original study by Denise et al, we classified patients into seven groups (Supplementary Figure 29A). We conducted a detailed analysis correlating tumor subtypes with patient drug responses, yielding quantifiable insights into the efficacy of treatment across different patient subgroups. As this dataset lacked patient survival information, we utilized immune infiltration scores and TIDE immunotherapy predictions, in addition to the aforementioned drug response patterns, as quantitative metrics (Supplementary Figures 29A, 26B).

Subsequently, we applied the existing subtyping methods available on our platform to molecularly classify Denise et al.'s data. Our quantitative analysis spanned ten drug treatments, immune infiltration scores, and TIDE immunotherapy predictions, allowing for a comprehensive comparison and visualization of the effectiveness of each subtyping method. We visualized the p values for each method's quantitative results. For all drug treatments (All arms), the results of all methods were statistically significant ($P < 0.05$, ANOVA test). The top three methods with the most significant p values were Denise et al., Immune-Cancer Pathway, and Stemness (Figure 6A). In terms of individual drugs, it was found that Denise et al., Stemness, and Immune-Cancer Pathway methods demonstrated a certain degree of stability, with statistical significance observed in over half of the quantitative metrics (Figure 6A). Regarding the quantization of the TME status (Figure 6A), Denise et al., Metabolism, TME2, and Cancer Pathway methods all yielded statistically significant results ($P < 0.05$, ANOVA test). In terms of TIDE immunotherapy prediction, all methods provided statistically significant results ($P < 0.05$, ANOVA test), with the top three methods being TME2, Denise et al., and TME1 (Figure 6A). The radar chart results revealed that the top three subtyping methods for overall performance were Denise et al., Immune-Cancer Pathway, and Stemness (Figure 6B, 6C). Additionally, we present the ranking of the statistical significance for different subtyping methods in each drug prediction metric (Figure 6D). Our findings highlighted Denise et al.'s method as particularly effective across all drug (All arm) treatments, which confirmed the findings reported in the original study. Notably, for the specific drug AMG386, the Immune-Cancer Pathway method emerged as the most suitable choice, illustrating how Themis

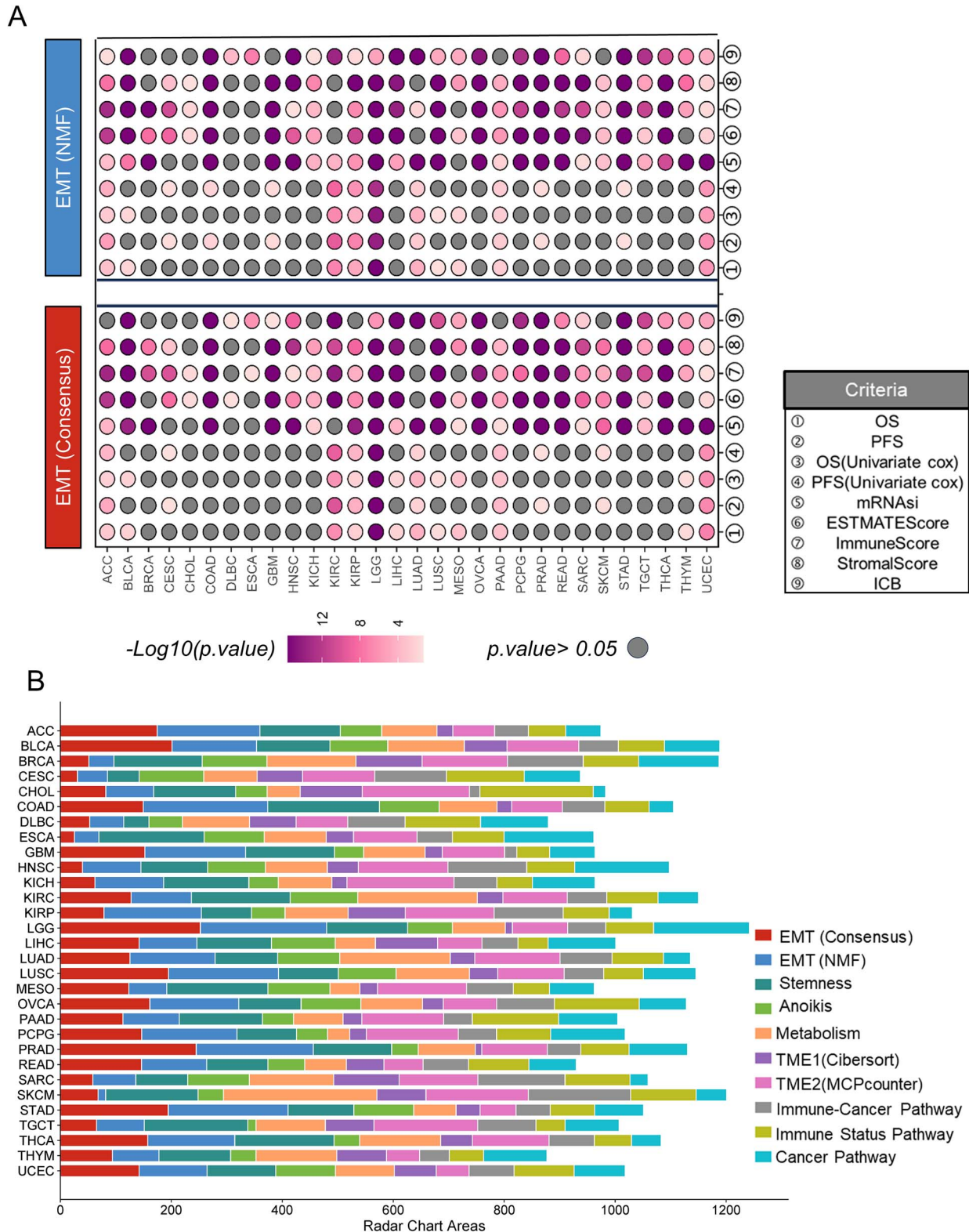


Figure 5. Performance of EMT subtyping method in pan-cancer. (A) We visualized the EMT (consensus) and EMT (NMF) methods' performance in terms of pan-cancer availability. For all colored points, darker shading indicates better performance and grey points represent $p > 0.05$. (B) the bar chart displays a quantitative comparison of the radar plot area between the quantitative results of the EMT (NMF/consensus) method and other methods in the datasets of 30 different tumor types in TCGA.

can guide personalized medical strategies (Figure 6D). Users can select a subtyping method based on their individual needs for personalized medical strategies.

Discussion

In precision oncology, research on tumor molecular subtyping plays a crucial role in the personalized management of patients

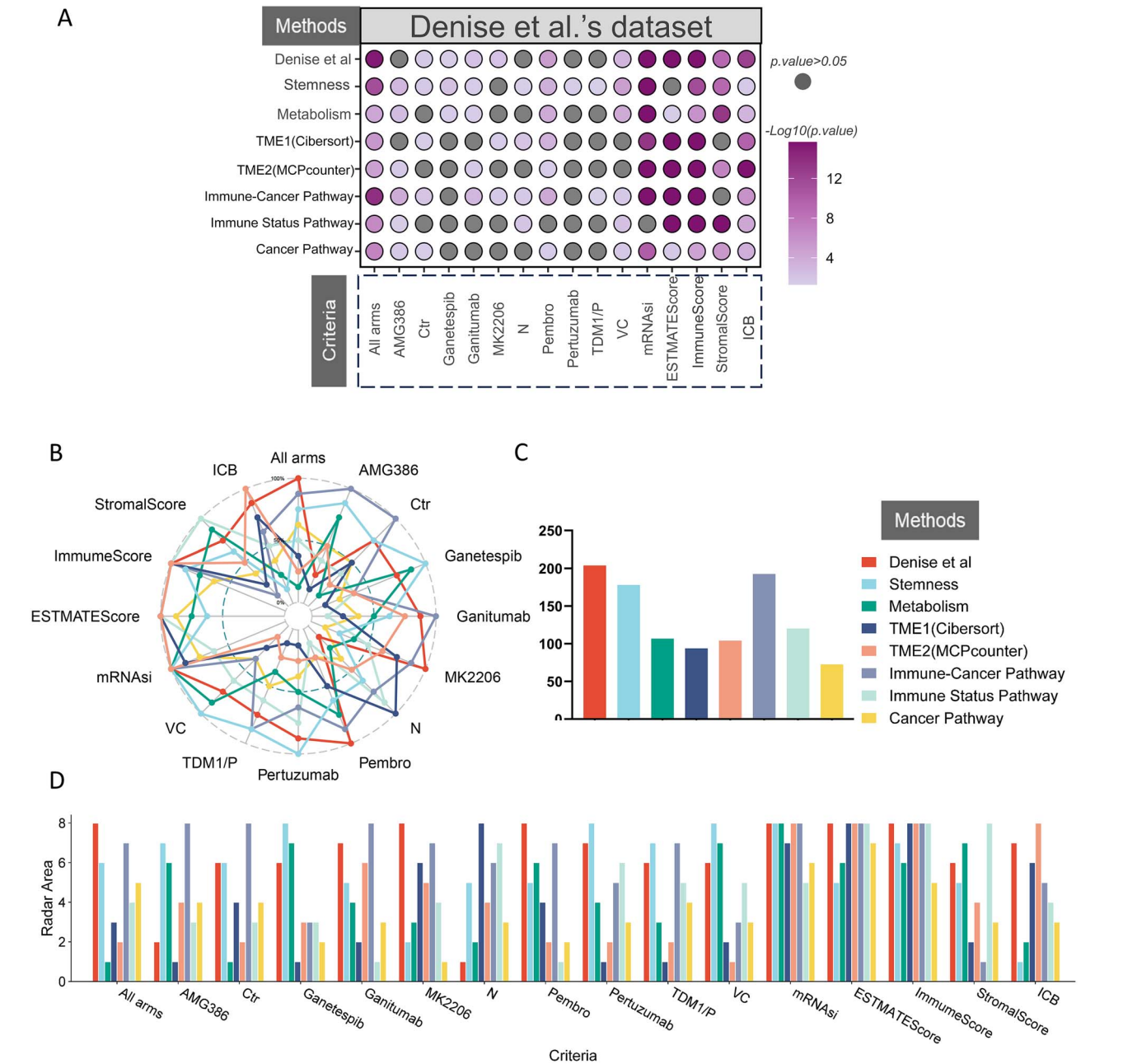


Figure 6. **Example data Denise et al.'s application.** (A) We visualized the performance of each subtyping method in the Denise et al data. For all colored points, darker shading indicates better performance and grey points indicate the $p > 0.05$. (B) Use of radar charts to visualize the performance of each subtyping method in the harmonized quantitative metrics. (C) Bar graph showing the quantitative comparison of radargram area for all methods. (D) Bar chart visualizing the statistical significance ranking of quantitative results presenting different subtyping methods in different indicators.

in clinical practice [34]. This study presents a benchmark investigation and strategic usage framework for tumor molecular subtyping methods, culminating in the development of Themis—a platform encapsulating all relevant analysis methods. We summarized eight representative tumor subtyping methods, along with associated test datasets, as the foundation for our work. We evaluated these methods using eight test datasets and TCGA tumor datasets, employing multiple metrics to assess their robustness, and pan-cancer applicability. Quantitative analyses were performed using four clinical metrics, including survival prognosis, immune infiltration scores, mRNAsi scores, and treatment predictions, to compare the associations between patient subtypes obtained from different methods and clinical

phenotypes. Subsequently, we visualized the p values of the quantitative results and conducted a comprehensive performance comparison using radar charts and radar chart areas. We found that the established subtyping methods exhibited varying performances in different application scenarios. This highlights the need to optimize tumor subtyping methods based on specific cases. For instance, by assessing the overall performance of each method, we observed that in the ACC dataset, the top-performing methods were Stemness, Metabolism, and TME2. Therefore, the Stemness method may be a suitable choice for application in the ACC dataset. Furthermore, we explored the possibility of combining the top two methods, Stemness and Metabolism, from the ACC results to propose a novel subtyping method,

Stemness-Metabolism, which outperformed both Stemness and Metabolism individually. Themis offers the potential for optimizing existing methods and evaluating users' methods and data, thereby providing personalized tumor subtyping strategies.

In the quantitative analysis conducted on the Themis platform, a set of clinical indicators was utilized for comparison. Specifically, the most clinically relevant OS and PFS were employed to conduct correlation-based quantitative analysis around the patient's survival prognosis. This approach holds significant value, allowing researchers to comprehensively evaluate the survival prognosis of patients, reveal the effectiveness of treatment plans, and track the disease progression of patients. Furthermore, deploying the ESTIMATE R package [31] was a widely accepted method that facilitates a comprehensive understanding of the patient's immune microenvironment status. It held great significance in predicting treatment outcomes, assessing immunotherapy responses, and guiding tailored treatment decisions. Treatment prediction encompassed two aspects: immunotherapy prediction using the TIDE algorithm [32] and chemotherapy drug prediction using the pRRopetic R package [33]. Currently, there are other immunotherapy prediction methods that use the easier R package [35], submap algorithm [36], or ImmuneCellAI algorithm [37]. Immunotherapy prediction methods are also available with Precily R package [38], oncoPredict [39], CellMiner [40] or DrugComb [41]. The TIDE algorithm [32] and pRRopetic R package [33] were adopted in the current version of Themis due to their high citation rates, which provided a certain degree of reliability and scientificity for treatment prediction. As an open platform, Themis provides great flexibility, making it relatively easy to constantly change and optimize around quantitative methods.

In applying the Themis platform, we leveraged a variety of subtyping criteria and clustering methods, reflecting the inherent complexity and diversity of tumor molecular subtyping. When conducting tumor subtype classification research, tumor heterogeneity is a key factor [42]. Tumor heterogeneity, characterized by biological variances among different patients and tumor types, presents a significant challenge in cancer classification and treatment [43]. Due to the intra-heterogeneity of tumors, the same type of cancer may have different subtype classifications, which differ in gene expression, mutation patterns, and biological characteristics [44, 45]. Therefore, when conducting tumor subtype research, it is necessary to carefully consider subtyping criteria and clustering methods that are suitable for specific cancer types. To ensure the stability and repeatability of subtypes, it is necessary to accumulate and validate large-scale data and incorporate diverse data from different time points, sources, and populations [46]. In addition, different tumor types may require different subtype classification methods to answer specific clinical questions [47]. For example, based on results from Themis, Stemness subtyping Method might be suitable to predict BLCA patient's drug response (ICB), and BLCA patients with high stemness showed better immunotherapy efficacy (Supplementary Figure 30A), which is consistent our previous study [48]. However, TME1 subtyping Method outperformed the other methods in predicting the patient's OS and PFS (Supplementary Figure 30B). Therefore, the Themis platform can help users comprehensively consider tumor intra-heterogeneity and clinical application needs and compare and validate appropriate subtyping methods. This is an ongoing process as new data and technologies continue to emerge, requiring continuous updates and improvements to the platform in the future.

In summary, the Themis platform revolves around the correlation between tumor patient subtyping and clinical characteristics, considering personalized patient subtyping needs in clinical practice. Looking ahead, a key challenge lies in expanding Themis's scope beyond tumor subtyping to encompass a wider array of diseases, thereby establishing a versatile and universally applicable platform.

Conclusions

Themis uniquely evaluates various subtyping methods using 38 datasets across 30 tumor types, providing a nuanced understanding of their effectiveness in different clinical contexts. A key innovation of Themis is its subtyping result quantification framework. This framework not only identifies the most suitable subtyping methods for specific clinical scenarios but also aids in developing new methods. The tool offers unique features and capabilities and is available at GitHub: <https://github.com/Hao-Zou-lab/Themis>.

Methods

Data acquisition and processing

Our test datasets were derived from these papers based on the classification of various subtyping methods: Wang et al. [13], Sun et al. [12], Yang et al. [11], Zeng et al. [22], Zheng et al. [21], Li et al. [17], Ye et al. [16], Cheong et al. [15]. Furthermore, we utilized a comprehensive, normalized data matrix from TCGA, encompassing 30 diverse tumor types [29], to supplement our analysis. All clinical information is summarized in Supplementary Tables S1–S2.

These eight papers contain data named sequentially with Data1 [13], Data2 [12], Data3 [11], Data4 [22], Data5 [21], Data6 [17], Data7 [16], and Data8 [15] and correspond to the subtyping methods Stemness [13], Anoikis [12], Metabolism [11], TME1 [22], TME2 [21], Immune-Cancer Pathway [17], Immune Status Pathway [16], and Cancer Pathway [15], respectively. The detailed processing of these eight test datasets can be found in the Supplementary Information File.

Strategic implementation of diverse tumor subtyping methods

The eight tumor subtyping methods were systematically categorized based on their respective criteria: feature-based methods (Stemness [13], Anoikis [12], and Metabolism [11]), deconvolution-based cell composition pattern methods (TME1 [22] and TME2 [21]) and pathway-based methods (Immune-Cancer Pathway [17], Immune Status Pathway [16], and Cancer Pathway [15]). Immune-Cancer Pathway: Immune (natural killer cell-mediated cytotoxicity, antigen processing and presentation, T cell receptor signaling, B cell receptor signaling, and Fc gamma R-mediated phagocytosis), stromal (ECM-receptor interaction, focal adhesion, and tight junction), DNA damage repair (p53 signaling, mismatch repair, and homologous recombination), and oncogenic signatures (PI3K-Akt signaling, Wnt signaling, TGF- β signaling, and cell cycle). Immune Status Pathway: Antigen Processing and Presentation, Antimicrobials, BCR Signaling Pathway, Chemokine Receptors, Chemokines, Cytokine Receptors, Cytokines, Interferon Receptor, Interferons, Interleukins, Interleukins Receptor, Natural Killer Cell Cytotoxicity, TCR signaling Pathway, TGF β Family Member, TGF β Family Member Receptor, TNF Family Members, TNF Family Members Receptors. Cancer Pathway: DNA damage response, TGF- β signaling, and cell proliferation pathways. The specific

application of these methods can be found in the Supplementary Information File, Section 5.

Correlation between molecular subtyping results and clinical phenotypes

We evaluated the tumor subtyping methods based on their correlations with clinical phenotypes, encompassing survival prognosis, immune infiltration scores, mRNAsi scores, and treatment predictions (Supplementary Information File, Section 6). The credibility of associations is measured through statistical significance and represented by hypothesis testing methods and p values used in its calculation. Therefore, we aggregated p values from all methods and constructed visual summary tables to display the performance of these eight subtyping methods (Supplementary Information File, Section 7). Then, we ranked the p values of the eight subtyping methods based on the quantitative results in descending order, with smaller p values receiving higher ranks. We then aggregated the ranking results for the eight datasets and the pan-cancer datasets separately. We used a radar chart to compare the ability measured by multiple dimensions, which are composed of rank quantities (Supplementary Information File, Section 8).

Optimizing tumor subtyping methods for enhanced precision

To optimize subtyping approaches, we integrated the top-performing methods for ACC, specifically, Stemness and Metabolism as demonstrated in Figure 4F. This methodology stratifies ACC patients into four subgroups defined by varying levels of stemness and metabolic activity: High stemness-Low metabolism, High stemness-High metabolism, Low stemness-High metabolism, and Low stemness-Low metabolism.

Statistical analysis

All computational calculations and statistical analyses were conducted using the R programming language, ensuring a robust and reproducible analytical framework. Numeric variables were compared using t-tests or ANOVA test, while categorical variables were compared using chi-square tests, Fisher's exact test, or the Kruskal-Wallis test. Survival differences were compared using the log-rank test. Confidence intervals (CIs) were reported as 95% and significance was set at $P < 0.05$.

Key Points

- Themis is the first platform to provide comprehensive evaluation services for tumor molecular subtyping methods. Users can add their own methods for performance evaluation, guiding them to customize subtyping methods suitable for specific clinical scenarios.
- Themis comprehensively evaluates the effectiveness of different subtyping methods by quantitatively analyzing clinical indicators such as survival prognosis, immune infiltration score, and treatment prediction.
- Themis systematically applied on TCGA's dataset of 30 different tumor types, tailoring classification methods for different tumor types.

Supplementary data

Supplementary data is available online at *Briefings in Bioinformatics* online.

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Data availability

The datasets supporting the conclusions of this article are included within the article and its additional files. A summary of the data is shown in the Supplementary Table. All other relevant data supporting the key findings of this study can be obtained in the article and its Supplementary Information File or as reasonably requested from corresponding authors. We have developed an R package called Themis, which provides a unified interface for various tumor subtyping methods. This software package is freely available on GitHub: <https://github.com/Hao-Zou-lab/Themis>. The raw data that support the findings of this study are available on request from the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Authors' contributions

Jie Hao (Conceptualization, Methodology, Project administration, Writing—original draft), Yiming Zhang (Conceptualization, Project administration, Writing—original draft), Xin Zou (Conceptualization, Methodology, Project administration, Writing—original draft), Fulan Deng (Methodology, Writing—original draft), Kun Zheng (Methodology, Investigation, Writing—original draft), Yue Xi (Investigation, Writing—original draft), Yujun Liu (Writing—original draft), Wantao Chen (Writing—original draft, clinical consulting), Yingchun Zhang (Writing—original draft), Hourong Sun (Clinical consulting), Yingxia Zheng (Clinical consulting), Henry H. Y. Tong (Clinical consulting), and Yuan Ji (Clinical consulting). All authors are involved in the discussion and finalization of the manuscript.

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